

Quantitative Asset Analysis: Gold, S&P 500 & IBEX 35

Financial Data Science Case Study: Volatility assessment, cross-asset correlation, and risk-adjusted efficiency (Sharpe Ratio).

Executive Summary

This study analyzes the historical daily performance, volatility, and risk-adjusted efficiency of three major market benchmarks (**Gold**, **S&P 500**, and **IBEX 35**) to evaluate capital efficiency beyond gross returns.

Key Findings

- **Efficiency Leader: IBEX 35** achieved the highest Sharpe Ratio (**1.76**), offering the highest excess return per unit of total risk.
- **Gold's Volatility:** Gold registered the highest daily volatility ($\sigma = 1.59\%$), acting as a high-volatility asset over the analyzed timeframe.
- **Effective Diversification:** A low cross-asset correlation ($r = 0.12$) between Gold and the S&P 500 confirms Gold's role as an effective short-term portfolio diversifier.

Quantitative Metrics Summary

Asset	Daily Volatility (σ)	Annualized Sharpe Ratio ($R_f = 3\%$)	Rating
IBEX 35	1.07%	1.76	Excellent
Gold	1.59%	1.05	Acceptable
S&P 500	1.06%	0.81	Insufficient

Quantitative Methodology

Annualized Sharpe Ratio

Formulated by William F. Sharpe (1966), the Sharpe Ratio measures excess return per unit of total risk:

$$\text{Annualized Sharpe Ratio} = \frac{\bar{R}_p - R_{f,\text{daily}}}{\sigma_p} \times \sqrt{252}$$

Where:

- $\bar{R}_p$: Mean daily return of the asset.
- $R_{f,\text{daily}}$: Fixed daily risk-free rate derived from a 3% annual benchmark ($0.03/252$).
- σ_p : Standard deviation of daily returns.
- $\sqrt{252}$: Annualization factor based on 252 trading days.

Tech Stack & Dependencies

- **Language:** Python
- **Data Extraction:** `yfinance`

- **Data Processing:** `pandas`, `numpy`
- **Visualization:** `matplotlib`, `seaborn`

Visual Analysis & Market Data

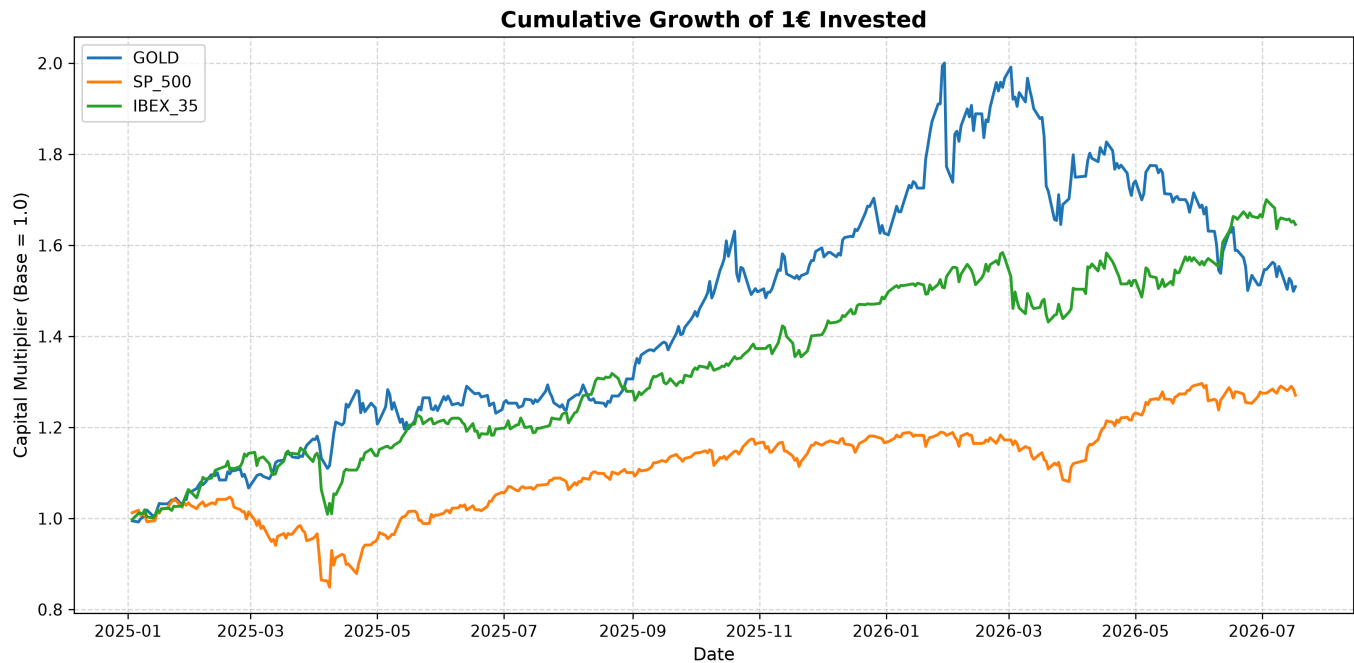
1. Daily Percentage Returns Statistics

Summary statistics for daily percentage returns (%) across the analyzed timeframe:

Metric	Gold (GC=F)	S&P 500 (^GSPC)	IBEX 35 (^IBEX)
Data Count	397	397	397
Mean Daily Return	0.116%	0.066%	0.131%
Daily Volatility (σ)	1.588%	1.065%	1.075%
Best Day (Max)	6.083%	9.515%	4.323%
Worst Day (Min)	-11.366%	-5.975%	-5.831%

2. Cumulative Growth of 1€ Invested

Evolution of historical capital performance over time:



3. Cross-Asset Correlation Matrix

Evaluating diversification benefits using daily return correlation (r):

Asset	Gold	S&P 500	IBEX 35
Gold	1.00	0.12	0.21
S&P 500	0.12	1.00	0.29
IBEX 35	0.21	0.29	1.00

Takeaway: The low correlation of **0.12** between Gold and the S&P 500 confirms that Gold acts as an effective short-term risk diversifier.

4. Risk-Adjusted Performance (Annualized Sharpe Ratio)

Comparing excess return relative to total risk ($R_f = 3\%$):

